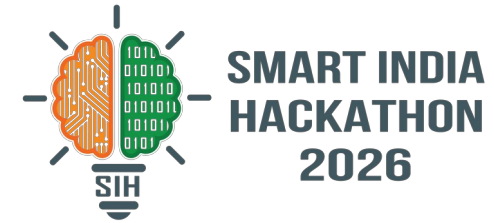


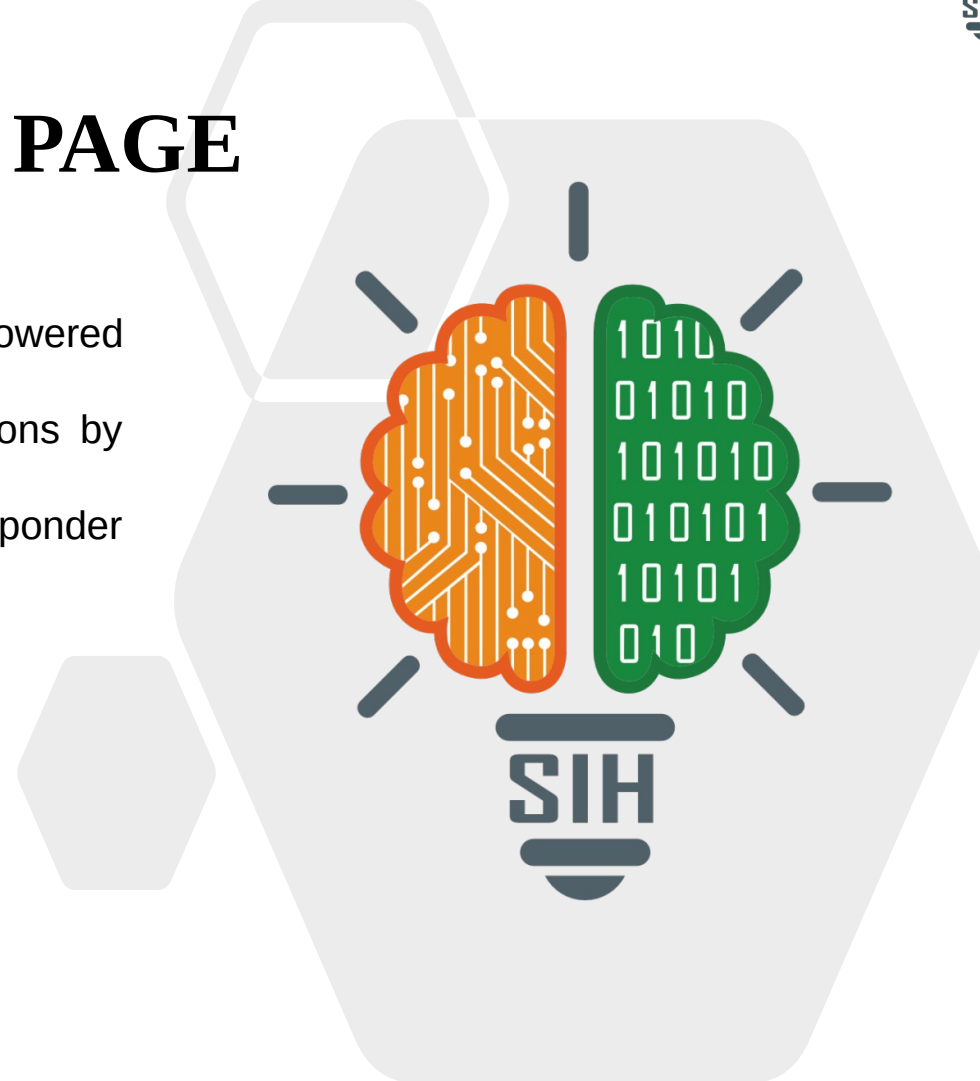
SMART INDIA HACKATHON 2026



TITLE PAGE

- **Problem Statement ID – 26177**
- **Problem Statement Title –** A deployable AI-powered autonomous drone that aids search-and-rescue operations by detecting people and hazards, thereby improving responder safety and reducing victim discovery time.
- **Theme – Robotics and Drones**
- **PS Category – Hardware**
- **Team ID –**
- **Team Name (Registered on portal) – ARES**

Live site ares.paralux.in Source github.com/ARES-UAV/ARES
Engineering handbook github.com/ARES-UAV/ARES/docs/handbook



[Watch: the full pitch](#)

IDEA/SOLUTION:

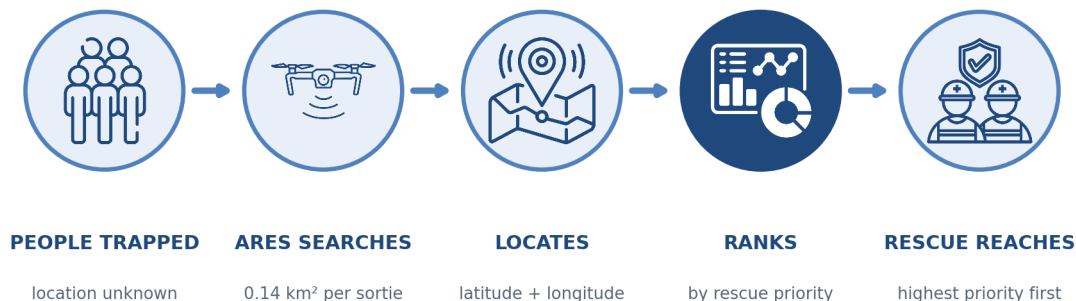
- **The problem:** After an earthquake or flood, the first hours decide who survives. Search teams fly fixed patterns a lawnmower grid that treats every square metre as equally likely to hold someone. It is not.
- **Why it matters:** Survival odds fall sharply with every hour under rubble. Time over empty ground is time taken from ground that holds someone.
- **Our idea:** A UAV that treats search as a decision, not coverage. Onboard AI finds survivors, ranks them, and re-plans its own path in the air with the radio switched off.

Problem Resolution:

- A UAV that decides where to look next, instead of covering ground in a fixed order.
- Onboard AI detects survivors in RGB and thermal, counts each person once, places them on a map and ranks them by rescue priority.
- A belief map of where people are likely to be is updated every time the drone looks somewhere — and the planner flies to whichever cell is worth the most per second of flight.
- It re-plans its own path in the air, with the radio switched off.

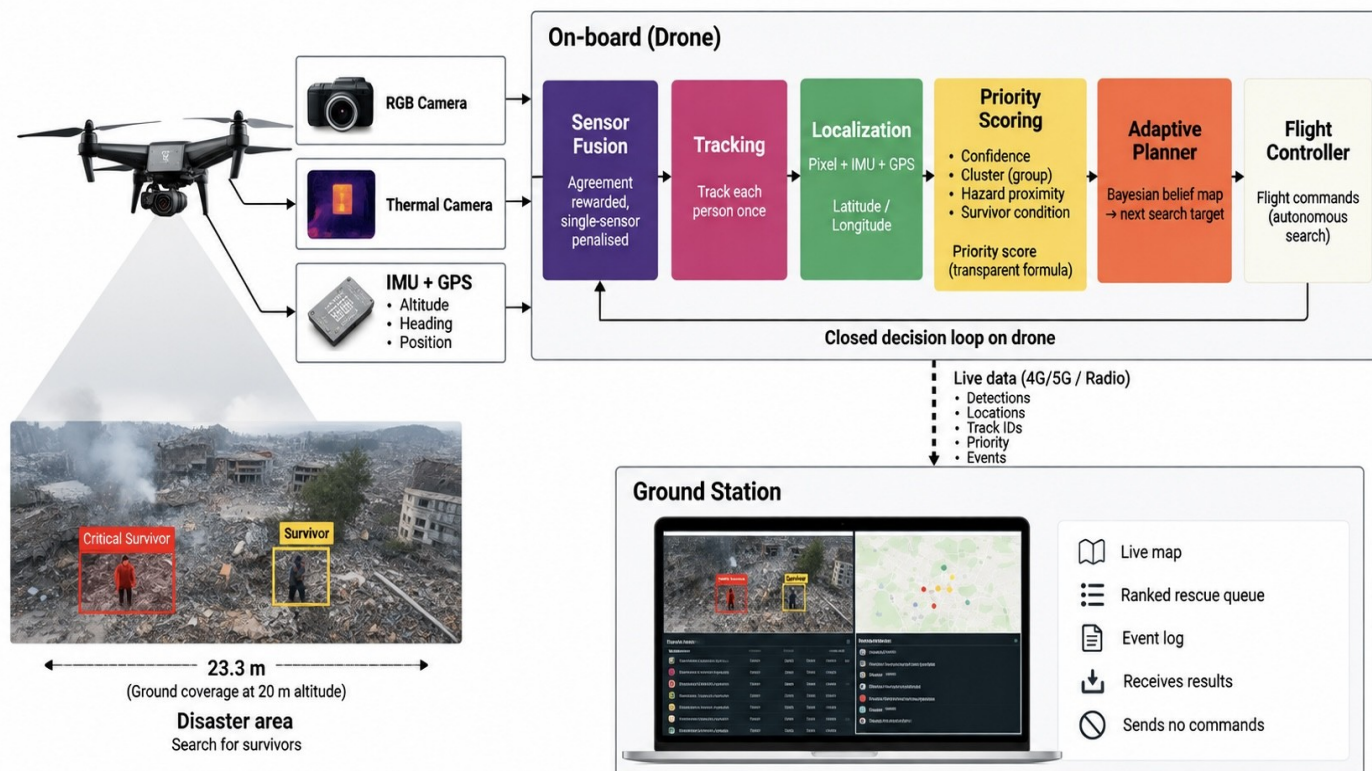
Unique Value Propositions:

- **Search as a decision, not coverage.** Priority changes the order of the search, not just the report at the end.
- **Nothing waits on a radio.** Every decision is made on the aircraft; the link carries information out, not commands in.
- **Explainable ranking.** A transparent weighted formula an officer can read — not a black box.
- **Honest scope.** We label what is measured, what is simulated, and what is only described.



The drone does not just find people — it decides who to reach first.

TECHNICAL APPROACH



NOT IN THE DIAGRAM

The models. YOLO, one person class — C2A + VisDrone for RGB, HIT-UAV for thermal.

The silicon. Qualcomm RB3 Gen 2, INT8 on the NPU: 209.5 ms a frame, 4.8 FPS measured.

Why it fits. Detection is the whole cost. Localization, priority and the planner together are 0.87% of the frame; the planner runs once per 20 s in 0.28 ms.

Why onboard. 60 decisions a sortie. On the ground each one is a radio round trip — 60 single points of failure.

Why the link is cheap. Records, not video: ~22 kbps against ~3,000. The whole mission is 2.7 KB.

Why 4.8 FPS is enough. At 20 m the footprint is 23 m and takes 7.7 s to cross — the same ground appears in roughly 37 frames.

Counted once. 7,081 raw detections become 333 confirmed tracks and 23 people.

What is simulated. Everything up to priority is measured. The planner has flown 30 simulated missions and no real ones.

Detection

YOLO · C2A + VisDrone · recall 0.824

Thermal

YOLOv8s · recall 0.883 · night and smoke

Tracking

ByteTrack · 7,081 detections → 23 people

Adaptive planner

Bayesian belief map · 0.28 ms per decision

Edge compute

Qualcomm RB3 Gen 2 · INT8 · 4.8 FPS

Flight stack

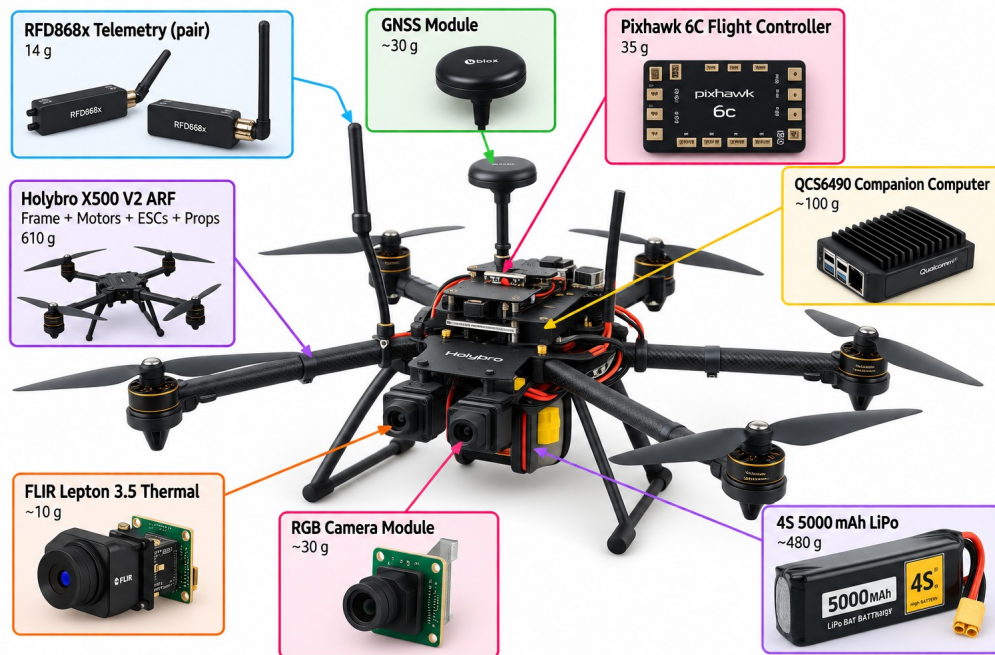
Pixhawk-class · ArduPilot · MAVLink

Backend / Frontend

FastAPI · React · Leaflet

Downlink

865–868 MHz · licence-exempt in India



ONE FRAME BECOMES ONE DECISION

[▶ Watch: the live dashboard](#)

frames

7,081

333

23

1

raw video

detections

confirmed

people

target

The narrowing is the argument: each person is counted once.

SOFTWARE STACK

Detection · tracking

Ultralytics YOLO + ByteTrack

AGPL-3.0

Flight stack

ArduPilot over MAVLink

GPL-3.0

Backend

FastAPI (Python)

MIT

Dashboard

React · Tailwind · Leaflet

MIT / BSD

Maps

OpenStreetMap tiles

no API key

Quantisation

Qualcomm AI Hub · INT8

free tier

Licence cost: zero.

IS IT VIABLE IN THE FIELD?

0.14 km² per sortie. A 23.3 m swath at 20 m altitude, 5 m/s, twenty minutes. Two sorties cover a village block.

DGCA Micro class. At ~1.4 kg it stays under the 2 kg ceiling: registration and a remote pilot certificate, no type certification.

No infrastructure. No tower, no internet, no ground station needed to fly. The link is optional by design.

One operator, one backpack. Under 2 kg all-up, assembled in minutes, launched from any clear ground.

Endurance is the real constraint. Holybro rates this airframe at ~18 min hover unloaded on 5 Ah; our payload cuts into that. A larger battery buys time but pushes takeoff weight past 2 kg into DGCA Small.

~1.4 kg

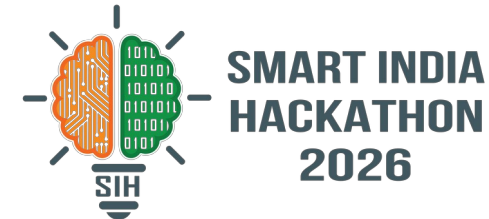
all-up takeoff weight — inside DGCA's Micro class, under the 2 kg ceiling. Registration and a remote pilot certificate; no type certification.

WHAT THIS DOES NOT YET PROVE

No aircraft has been built — this is a specified bill of materials, not a flown platform. The 209.5 ms benchmark ran on the RB3 dev kit on a bench supply. RGB and thermal together measured 2.4 FPS, not 4.8. In-flight power draw and thermal behaviour are unmeasured.

ARES

IMPACT AND BENEFITS



Finds more people

20 of 20 survivors against a lawnmower grid's 11.

Finds them sooner

Half the survivors reached 3.1× faster.

Works with no network

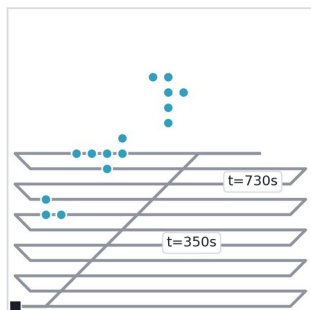
The full mission report is 2.7 KB.

Ranks who to reach first

Confidence, cluster size, hazard proximity.

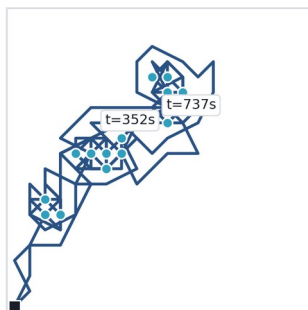
▶ [Watch: adaptive planner, 100 seeds](#)

FIXED PATTERN

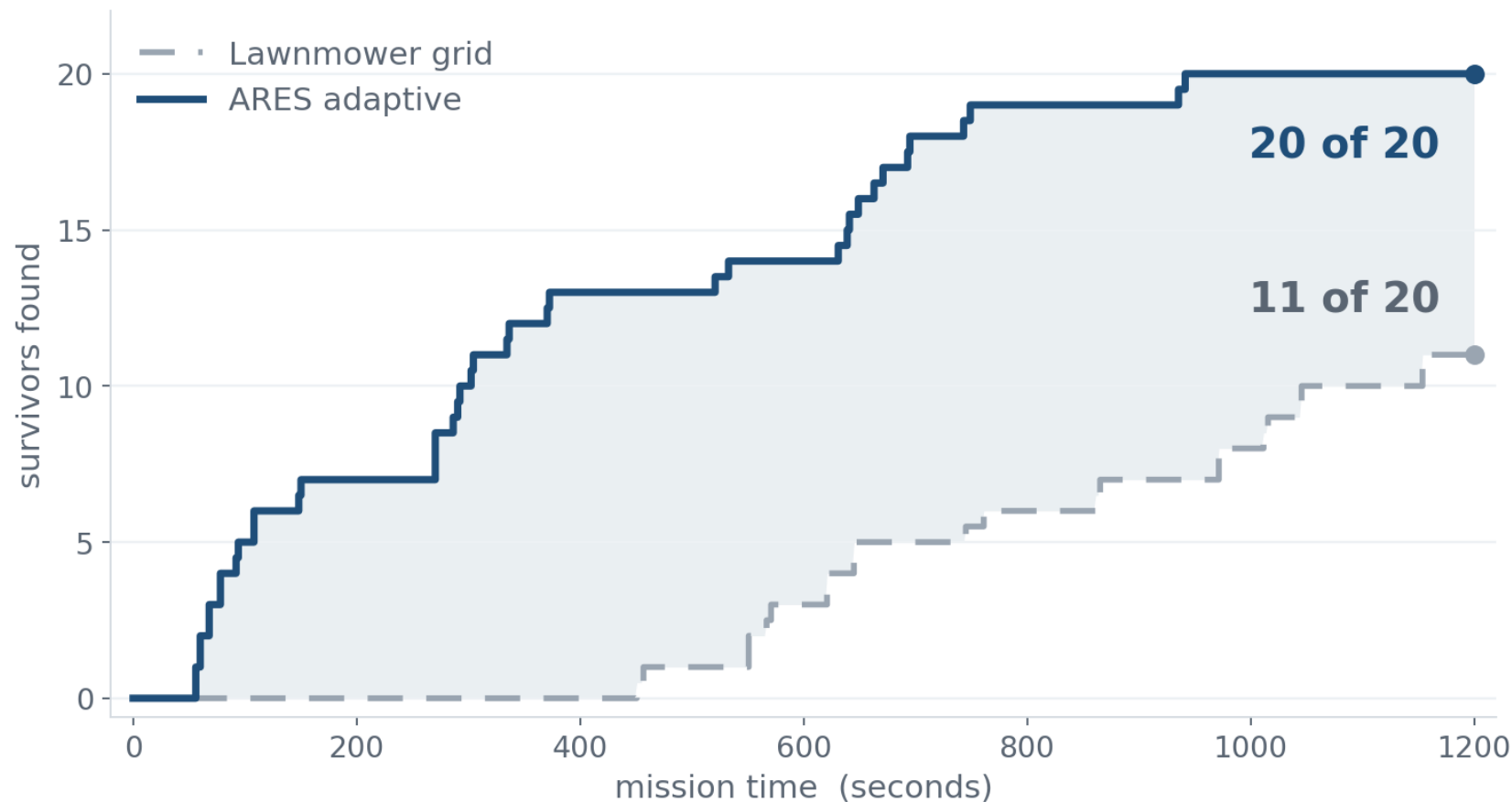


10 of 20 found

ARES ADAPTIVE



20 of 20 found



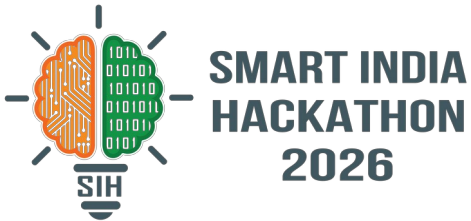
Survivors found over time — median of 100 simulated missions, good prior.

Simulated flight with the detector at our measured 0.824 recall. Adaptive covers 32% of the area against the grid's 56% — more people, sooner, over less ground. Never flown on hardware.



RESEARCH AND REFERENCES

LIVE SITE ares.paralux.in SOURCE CODE github.com/ARES-UAV/ARES ENGINEERING HANDBOOK [13 chapters](#)



C2A

disaster-scene humans

[LINK](#)

VisDrone

real aerial capture

[LINK](#)

AIDER

UAV hazard imagery

[LINK](#)

Ultralytics

detection + export

[LINK](#)

ByteTrack

stops double-counting

[LINK](#)

Qualcomm AI Hub

INT8 benchmarking

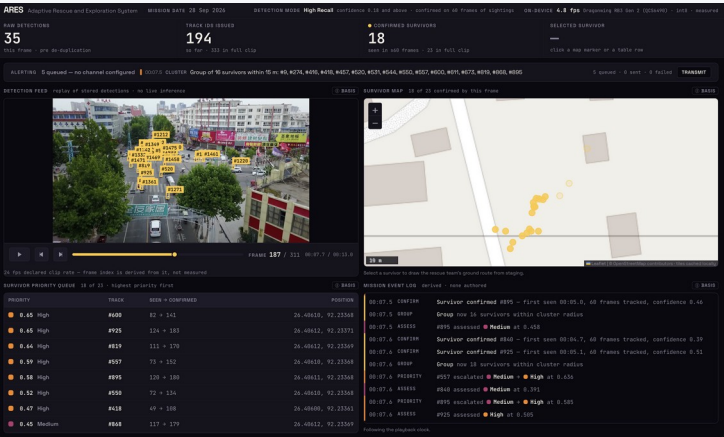
[LINK](#)

WPC 865–868

India licence-exempt band

[LINK](#)

THE DASHBOARD, RUNNING



18 of 23 confirmed at frame 187. Counts reconcile across every panel.

VIDEO EVIDENCE

Every claim here has a recording behind it.

[01 • The case for ARES](#)
the system, and what we did not build

[02 • Adaptive planner](#)
100 missions, plus the no-information control

[03 • Detection to GPS](#)
7,081 boxes become 23 people

[04 • Command dashboard](#)
live map, ranked queue, alerts held offline

WHAT IS BUILT

Detection, tracking, localization	BUILT
Priority scoring	BUILT
Command dashboard	BUILT
On-device benchmark	MEASURED
RGB + thermal fusion	PARTIAL
Hazard classification	PARTIAL
Adaptive planner	SIMULATED
GPS-denied SLAM	DESCRIBED